

30/07/26.

ITA0502 - ACTIVITY

SCENARIO BASED ACTIVITY

1. Technique Used: Median Filter.

Reason: Median filtering effectively removes salt & pepper noise while preserving image edges & details.

2. Technique Used: Gaussian Filter

Reason: Gaussian filter smooths the images and reduces Gaussian noise produced by imaging sensors.

3. Technique Used: Bilateral Filter

Reason: Bilateral filter removes noise while preserving important edges, making it suitable for a medical images.

4. Technique Used: Median Filter

Reason: Median filtering removes isolated black and white dots without affecting the document's text.

5. Technique Used: Gaussian filtering

Reason: Gaussian filtering reduces grainy noise & produces a smoother image when edge preservation is not critical.

6. Technique Used: Histogram Equalization

Reason: Histogram equalization enhances contrast, making roads and objects more visible in foggy conditions.

7. Technique Used: Contrast Stretching

Reason: Contrast stretching expands the intensity range, making underwater images brighter and more vivid.

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Techniques Used: Adaptive histogram Equalization

Reason: CLAHE improve local contrast, making bones and tissues more visible without over enhancing image.

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Techniques Used: Histogram equalization

Reason: Histogram equalization increase Brightness and contrast improving visibility in low-light traffic images.

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Techniques Used: Contrast stretching

Reason: Contrast stretching expands the narrow intensity range to utilize the full grayscale spectrum improving image quality

